

Quenverell Computational Ltd, a Tarnovell company

Stress-kinase ligands: crystallography-based docking and pocket-water assessment

Final client report

Engagement: QV.25.086

Client: Drelvessa Therapeutics SAS

Date: 13/11/2025

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1 Question

Drelvessa's chemists asked where their stress-kinase series could be extended without losing the hinge interaction, and whether conserved water molecules in the ATP pocket should be displaced or kept.

2 Data provenance

- Three client-supplied X-ray crystal structures of the kinase domain with bound ligands (resolution between 1.8 and 2.3 Å)
- Electron density maps reviewed for ligand and water placement before use
- Client compound set of current analogues and proposed designs (SMILES), checked for tautomer and protonation states

Data acceptance: one structure was excluded from scoring because the ligand density did not support the deposited pose; it was used only for visual comparison.

3 Methods

- Protein preparation from the crystal structures with conserved waters retained or removed in separate runs
- Docking with hinge hydrogen-bond constraints; pose clustering and visual review with the client's chemists
- Pocket-water assessment: waters seen in all three structures treated as conserved; their displacement scored separately
- Ranking by consensus across the two retained crystal structures

Docking scores rank poses within this pocket only; they are not binding affinities and are not compared across chemotypes.

4 Findings

- The hinge-binding pose of the series is reproduced in both retained crystal structures.

- Two conserved waters bridge the ligand to the back pocket; designs that displace them lost the consensus pose and are not recommended.
- A solvent-facing vector from the 5-position is open in both structures and tolerates polar substituents.
- Designs extending towards the glycine-rich loop were ranked inconsistently between structures and are flagged as uncertain.

5 Recommendation

Prioritise analogues using the solvent-facing 5-position vector with small polar groups, keep the conserved waters, and confirm binding in the client's biochemical assay before any further structure-based work. If more confidence on the glycine-rich loop is needed, a new co-crystal structure with a representative analogue would resolve it.

6 Limitations

Conclusions rest on two accepted crystal structures of one conformational state. Induced-fit effects and protein flexibility were not modelled beyond side-chain refinement.

Name	Role	Signature	Date
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